Week 7

Case Study 1

Data for Questions 1 to 5

Water Demand Forecasting for a Residential Zone

A municipal analytics team builds models to predict next-day water demand for a residential zone. Each input contains temperature, weekday indicator, previous-day demand, and a local festival indicator. The team compares a simple linear model and a highly flexible neural network. The simple model gives similar predictions when trained on different samples, but its average prediction is far from the true demand. The neural network fits the training data very closely, but its predictions change substantially when trained on different samples.

For one input condition, the neural network predictions from four training subsets are

$$41, \quad 49, \quad 45, \quad 53$$

thousand liters, while the true demand is 48 thousand liters. The expected test error for this condition is decomposed as

$$\mathbb{E}\left[(y - \hat{f}(x))^2\right] = \text{Bias}^2 + \text{Variance} + \sigma^2.$$

Question 1: Which description best matches the simple linear model?

A) High bias and low variance

B) Low bias and high variance

C) Low bias and low variance

D) High variance and zero bias

Correct Answer: A

Explanation: Similar predictions across samples indicate low variance, while being far from the true demand indicates high bias.

Question 2: What is the average neural-network prediction for the given input?

A) 46

B) 48

C) 47

D) 50

Correct Answer: C

Explanation: The average is $(41 + 49 + 45 + 53)/4 = 188/4 = 47$.

Question 3: What is the squared bias of the neural-network predictions for this input?

A) 4

B) 9

C) 0

D) 1

Correct Answer: D

Explanation: The squared bias is $(48 - 47)^2 = 1$.

Question 4: What does the large variation among the neural-network predictions indicate?

A) The model has no irreducible error.

B) The model has high variance under the case assumptions.

C) The output labels are unavailable for the described model.

D) The model is too simple for the data.

Correct Answer: B

Explanation: Predictions changing strongly across training subsets are a symptom of high variance.

Question 5: Which action is most directly aimed at reducing the neural network's variance?

A) Add a suitable regularization method.

B) Remove the validation set entirely.

C) Increase model depth without constraints.

D) Train until training loss becomes zero.

Correct Answer: A

Explanation: Regularization methods constrain the model and help reduce overfitting-related variance.

Case Study 2

Data for Questions 6 to 10

Small-Business Loan Risk Scoring

A cooperative bank trains a logistic-regression model to predict whether a small-business loan application is risky. The model uses standardized features such as debt ratio, monthly revenue, repayment history score, and business age. Without regularization, the model reaches very high training accuracy but noticeably lower validation accuracy. Several learned weights also have large magnitudes, making the model sensitive to small feature changes.

The team adds L2 regularization to the objective:

$$J_{\text{total}}(w) = J_{\text{data}}(w) + \alpha \sum_i w_i^2.$$

For one parameter, the current value is $w = 3$. The data-loss gradient for that parameter is 4, the regularization strength is $\alpha = 2$, and the learning rate is $\eta = 0.1$. For this case, use the regularized gradient

$$\frac{\partial J_{\text{total}}}{\partial w} = 4 + 2\alpha w.$$

Question 6: What is the main purpose of adding L2 regularization here?

A) To remove the need for training data for the described model

B) To discourage large weights for the described model

C) To guarantee perfect validation accuracy

D) To convert logistic regression into clustering

Correct Answer: B

Explanation: L2 regularization penalizes large squared weights and reduces model complexity.

Question 7: What is the regularization contribution $2\alpha w$ to the gradient?

A) 10

B) 6

C) 8

D) 12

Correct Answer: D

Explanation: The contribution is $2\alpha w = 2(2)(3) = 12$.

Question 8: What is the full regularized gradient for this parameter?

A) 16

B) 12

C) 10

D) 14

Correct Answer: A

Explanation: The full gradient is $4 + 12 = 16$.

Question 9: What is the updated value of w after one gradient-descent step?

A) 2.0

B) 2.6

C) 1.4

D) 3.6

Correct Answer: C

Explanation: $w_{\text{new}} = 3 - 0.1(16) = 1.4$.

Question 10: Which outcome is expected after applying an appropriate L2 penalty?

A) Training accuracy must become zero for the data shown here.

B) Validation performance may improve due to reduced overfitting.

C) All feature weights must become exactly zero.

D) The model stops using gradient descent under the given conditions.

Correct Answer: B

Explanation: L2 regularization can reduce overfitting and improve validation performance, even if training accuracy decreases slightly.

Case Study 3

Data for Questions 11 to 15

Energy Consumption Prediction for an Office Campus

A facility-management company trains a neural network to predict hourly electricity consumption in an office campus. Inputs include outside temperature, hour of day, day type, and previous-hour load. The model is trained for many epochs, and both training and validation loss are monitored after every five epochs.

Epoch	Training Loss	Validation Loss
5	0.62	0.70
10	0.44	0.52
15	0.31	0.43
20	0.22	0.46
25	0.16	0.51

The team uses early stopping because the deployed model must generalize to future days, not only memorize historical operating patterns.

Question 11: At which epoch is the validation loss minimum?

A) Epoch 10

B) Epoch 20

C) Epoch 15

D) Epoch 25

Correct Answer: C

Explanation: The smallest validation loss is 0.43, which occurs at epoch 15.

Question 12: What pattern after epoch 15 suggests overfitting?

A) Both losses increase together.

B) Training loss increases while validation loss decreases.

C) Training loss becomes exactly zero.

D) Training loss decreases while validation loss increases.

Correct Answer: D

Explanation: After epoch 15, training loss continues decreasing but validation loss increases, indicating worsening generalization.

Question 13: Why should early stopping monitor validation loss rather than only training loss?

A) Validation loss reflects performance on unseen data.

B) Training loss is not computed during neural-network training.

C) Validation loss is always exactly equal to test loss.

D) Training loss cannot decrease across epochs.

Correct Answer: A

Explanation: Validation loss is used as a proxy for generalization to data not used for weight updates.

Question 14: Which statement about early stopping is correct?

A) It guarantees the globally optimal weights.

B) It acts as a regularization method by limiting over-training.

C) It prevents all forms of noise in input data.

D) It removes the need for a validation set under the case assumptions.

Correct Answer: B

Explanation: Early stopping limits over-training and therefore acts as a regularization method.

Question 15: If the team keeps the model from epoch 25 instead of epoch 15, what is the main risk?

A) The training set becomes larger automatically.

B) The model cannot produce numerical predictions.

C) The validation loss is ignored by definition under the case assumptions.

D) The model may generalize worse despite lower training loss.

Correct Answer: D

Explanation: Selecting epoch 25 may overfit the training data and generalize worse than the epoch with minimum validation loss.

Case Study 4

Data for Questions 16 to 20

Product-Surface Image Quality Inspection

A quality-control team trains a CNN to classify product-surface images into acceptable and defective classes. Only 900 labeled images are available. The training split is 70validation split is 30validation accuracy. The team applies image augmentation during training using small rotations, horizontal shifts, brightness changes, and mild zooms. Each original training image generates two additional augmented images.

The team also adds dropout to a hidden layer. During training, each hidden unit is kept with probability $p = 0.8$. At test time, the full network is used with appropriate scaling. Finally, three independently trained CNNs are averaged to create an ensemble.

Question 16: How many original images are in the training split before augmentation?

A) 270

B) 900

C) 630

D) 450

Correct Answer: C

Explanation: The training split contains $0.70 \times 900 = 630$ images.

Question 17: What is the effective training-set size after adding two augmented images per original training image?

A) 1890

B) 1260

C) 630

D) 1530

Correct Answer: A

Explanation: Each original contributes one original plus two augmented versions, so $630 \times 3 = 1890$.

Question 18: What is the main reason augmentation can improve validation accuracy?

A) It changes all class labels randomly in the stated case setting.

B) It removes the need to learn features.

C) It guarantees that training loss is zero.

D) It exposes the model to realistic input variations.

Correct Answer: D

Explanation: Augmentation improves robustness by exposing the model to realistic transformations of the inputs.

Question 19: During dropout training with keep probability $p = 0.8$, approximately how many units remain active in a layer of 50 hidden units?

A) 25

B) 10

C) 40

D) 50

Correct Answer: C

Explanation: The expected active units are $0.8 \times 50 = 40$.

Question 20: Why can averaging three independently trained CNNs improve reliability?

A) It increases the learning rate of each CNN.

B) It can reduce variance by combining diverse predictions.

C) It removes all bias from every CNN for the data shown here.

D) It makes the dataset labels unnecessary.

Correct Answer: B

Explanation: Averaging diverse models can cancel some individual errors and reduce prediction variance.